

MA.912.AR.3.8

Benchmark

MA.912.AR.3.8 Solve and graph mathematical and real-world problems that are modeled with quadratic functions. Interpret key features and determine constraints in terms of the context.

Algebra I Example: The value of a classic car produced in 1972 can be modeled by the function $V(t) = 19.25t^2 - 440t + 3500$, where t is the number of years since 1972. In what year does the car's value start to increase?

Benchmark Clarifications:

Clarification 1: Key features are limited to domain; range; intercepts; intervals where the function is increasing, decreasing, positive or negative; end behavior; vertex; and symmetry.

Clarification 2: Instruction includes the use of standard form, factored form and vertex form.

Clarification 3: Instruction includes representing the domain, range and constraints with inequality notation, interval notation or set-builder notation.

Clarification 4: Within the Algebra I course, notations for domain, range and constraints are limited to inequality and set-builder.

Connecting Benchmarks/Horizontal Alignment

- MA.912.AR.4.1
- MA.912.AR.5.3

Terms from the K-12 Glossary

- Coordinate Plane
- Domain
- Function Notation
- Quadratic Function
- Range
- x -intercept
- y -intercept

Vertical Alignment

Previous Benchmarks

- MA.8.AR.3.4

Next Benchmarks

- MA.912.AR.5.6

Purpose and Instructional Strategies

In grade 8, solved problems involving real-world linear equations. In Algebra I, solve problems that are modeled by quadratic functions. Students additionally graph the function and determine or interpret key features of the function. In later courses, this work expands to exponential and other kinds of functions.

- This benchmark is a culmination of MA.912.AR.3. Instruction here should feature a variety of real-world contexts. Some of these contexts should require students to create a function as a tool to determine requested information or should provide the graph or function that models the context.
- Instruction includes making connections to various forms of quadratic equations to show their equivalency. Students should understand and interpret when one form might be more useful than other depending on the context.
 - Standard Form
Can be described by the equation $y = ax^2 + bx + c$, where a , b and c are any rational number. This form can be useful when identifying the y -intercepts.
 - Factored Form
Can be described by the equation $y = a(x - r_1)(x - r_2)$, where r_1 and r_2 are real numbers and the roots, or x -intercepts. This form can be useful when identifying the x -intercepts, or roots.
 - Vertex Form
Can be described by the equation $y = a(x - h)^2 + k$, where the point (h, k) is the vertex. This form can be useful when identifying the vertex.
- Instruction includes the use of x - y notation and function notation.
- Instruction includes representing domain, range and constraints using words, inequality notation and set-builder notation.
 - Words
If the domain is all real numbers, it can be written as “all real numbers” or “any value of x , such that x is a real number.”
 - Inequality Notation
If the domain is all values of x greater than 2, it can be represented as $x > 2$.
 - Set-Builder Notation
If the domain is all values of x less than or equal to zero, it can be represented as $\{x|x \leq 0\}$ and is read as “all values of x such that x is less than or equal to zero.”
- Instruction provides opportunities to make connections between the domain and range and other key features.
 - For example, a coffee shop uses the function, $P(x) = -80x^2 + 480x - 540$ to model the profit they can earn in thousands of dollars in terms of the price per cup of coffee, in dollars. By determining the domain and range that includes prices that yield a positive profit, one would also have to identify the vertex (or maximum) and the roots of the function. Students should realize that they can do this by transforming the given expression into vertex form.

Common Misconceptions or Errors

- Students may find themselves stuck initially, unsure of where to start. In conversations with these students, prompt them to reflect on what they know about the context and how they can use that information to determine the requested information (*MTR.1.1*).
 - For example, students may have an equation in standard form and need to interpret the vertex in context. Prompting students to consider how they've calculated vertices in the past should lead them to choose to either convert the equation into vertex form or use the line of symmetry to help determine the vertex.

Strategies to Support Tiered Instruction

- Teacher provides equations in both function notation and x - y notation and models graphing both forms using a graphing tool or graphing software (*MTR.2.1*).
 - For example, $f(x) = (x - 2.3)^2 + 7$ and $y = (x - 2.3)^2 + 7$, to show that both $f(x)$ and y represent the same outputs of the function.
- Instruction provides opportunities to visualize the domain and range on a graph using a highlighter.
 - For example, a coffee shop uses the function $P(x) = -80x^2 + 480x - 540$ to model the profit they can earn in thousands of dollars in terms of the price per cup of coffee, in dollars. If the coffee shop is only interested in prices that yield a positive profit, the highlighted domain and range are shown.



Instructional Tasks

Instructional Task 1 (*MTR.2.1, MTR.5.1*)

ABC Pool Company is constructing a 17 feet by 11 feet rectangular pool. Along each side of the pool, they plan to construct a concrete sidewalk that has a constant width. The land parcel being used has a total area of 315 sq. ft. to construct the pool and sidewalks. The function $f(x) = 4x^2 + 56x - 128$, represents the situation described. Transform the function to determine and interpret its zeros.

Instructional Items

Instructional Item 1

A new coffee shop wants to maximize their profit within the first year of business. They determined the function, $P(x) = -80x^2 + 480x - 540$, models the profit they can earn in thousands of dollars in terms of the price per cup of coffee, in dollars. What price of coffee maximizes the coffee shop's profit?

Instructional Item 2

The value of a classic car produced in 1972 can be modeled by the function $V(t) = 19.25t^2 - 440t + 3500$, where t is the number of years since 1972.

Part A. In what year does the car's value start to increase?

Part B. Will the car's will ever begin to decrease? Assume it follows its modeled value.

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*